

G. W. Leibniz
ARS REPRESENTATORIA

Translated by David Jekel and Matthew McMillan

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Ars Representatoria

Difficile est aliquid toto suo genere novum dare. Hanc tamen artem talem esse negari non potest. Ea tanto praestantior est Algebra, quanto majus est figuras et motus fideliter et nativa
5 facie menti referre quam solos numeros seu magnitudines.

Ductu igitur hujus artis et ope notarum simplicissimarum omnem figuram, motum, machinam, totam denique naturam quatenus mechanice explicata habetur, in mente possumus depingere, sine figuris et modulis, idque tam exacte quam fig-
10 uris et modulis fieri posset. Immo multo exactius et utilius. Nam possumus ita paucis et exiguis lineis repraesentare, quod alio-
qui repraesentandum esset multis et magnis figuris. Et cum figurae atque – cum ad solida veniendum est – moduli multo
15 tempore ne dicam expensis indigeant, et motus per solas fig-
uras sine modulis difficillime repraesententur, possumus hac arte intra horam plures figuras et earum transmutationes sive motus describere, quam figurae delineari possent intra
20 septimanam, et moduli fabricari intra annum. Praesertim cum ex multis figuris et modulis pleraeque initio inquirendi
descriptae aut fabricatae comperiantur inutiles, tempusque et sumtus perdantur; hac vero delineatione et fabricatione mentali sive repraesentatoria, facile multae figurae multique motus exhibentur brevi tempore et nullo sumtu.

Representational Art

It is difficult to make something completely new in kind. Nonetheless, it cannot be denied that this art is such. It is as much superior to algebra as it is greater to render figures and motions in the mind faithfully and in their original aspect, than to do so only by numbers, i.e. by magnitudes.

Guided by this art, therefore, and with the help of the simplest signs, every figure, motion, machine, and in short the whole of nature (to the extent it is considered to be explained mechanically) we will be able to depict in the mind without figures and models, and just as precisely as could be done with figures and models. In fact, much more precisely and usefully. For we could thus represent with a few tiny lines what otherwise would need to be represented by large figures, and many of them. And whereas figures and (when it comes to solid bodies) models require such a lot of time, not to mention the expense, and motions are the most difficult things to represent by figures alone without models, with this art more figures and their transmutations (i.e. motions) could be described in a few hours than figures could be drawn in a week, or models built in a year. Particularly because most of the many figures and models described or built at the beginning of the investigation turn out to be useless, and the time and cost are lost forever; but by this mental (i.e. representational) drawing and building, many figures and many motions are easily produced in a short time at no cost.

At inquires idem facere possum imaginatione, ita sane aliqua ex parte. Sed ad hoc ipsum inventa est haec ars ut subleuet et perficiat imaginationem; *subleuet* ne forti illa ad figuras attentione quae sane in figuris et motibus implicatis persequendis
 5 maxima esse debet, nimis fatigetur; *perficiat* vero, quoniam imaginatio nostra non simul multa considerare potest, praesertim cum ea quae non simul videri possunt, sunt simul cogitanda, ubi phantasmata mentem deserint; hinc fit, ut
 10 distincte multa considerare; et per multas transmutationes ire, nihilque negligere aut transsilire, difficillimum, nec nisi a magnis artificibus in iis rebus in quibus jam multum habent usum fieri potest; quamquam et ipsi saepe peccent et haereant in difficilioribus.

15 Verum quod maximum est, praestat haec ars aliquid quod sit super omnem imaginationem, non tantum enim exhibet figuras sed et explicat atque exprimit, intimas earum naturas seu causas, adeoque rem reducit ad perfectam analysin et ex hac rursus resultantem synthesin.

20 Eaque ratione fit ut possimus omnes ordine modos enumerare efficiendi propositas figuras motusque; eligereque aptissima in rem praesentem, et determinare quod est impossibile, prorsus ut Algebra facit in numeris. Algebra hoc habet sane egregium, quod ejus ope etiam figurarum et motuum
 25 problemata possunt solvi, sed quia omnia a situ transferuntur ad magnitudines earumque habitudines seu ad calculum numerorum. Hinc ubi semel a figura rem transtulimus ad Algebram, jam imaginationem plane deserit mens, et quasi per metaphysica vagatur, dimensiones scilicet altiores quam
 30 ut in figuris dentur; unde ad exitum quidem pervenitur, sed miris ambagibus, a rei ipsius contemplatione remotis; omni

But you will say that I can do the same thing with the imagination. Yes of course, in part. But this art was invented for that very thing, namely to relieve and perfect the imagination. To *relieve* it lest it becomes excessively fatigued by close attention to figures, which of course is very great when pursuing complex figures and motions; and to *perfect* it, because our imagination cannot contemplate many things simultaneously, especially when things which cannot be simultaneously viewed are to be simultaneously thought, when the mental images [*phantasmata*] desert the mind. Hence it happens that before arriving at the
 5R last things, the first ones have slipped away. Certainly to contemplate many things distinctly, and to pass through many transmutations and neglect or skip over nothing, is quite difficult, and it cannot be done except by great artisans of these things who already have a lot of practice, although they themselves also slip up frequently and become
 10R entangled in difficulties.
 15R

But the best thing is that this art supplies something beyond imagining, for not only does it exhibit figures, but also it unfolds and expresses their inmost natures, or causes, and thus reduces the matter to a perfect analysis, and to a synthesis resulting from that in turn.

20R And it happens that by this method we could enumerate in order all the ways of making proposed figures and motions, and choose the most suitable for present purposes, and determine what is impossible, indeed just as algebra does with numbers. Algebra has of course the distinction that with its help we can also solve problems of figures and motions, but this is because everything is brought from situation over to magnitudes and their relationships, i.e. to calculation of numbers; hence, once we have brought the matter over from figure to algebra, the mind now completely deserts the imagination and it wanders as if through the metaphysical, that is, through higher dimensions than are given in figures; from here it reaches the way out, but by remarkable convolutions, removed far from contemplation of the thing itself, and

cogitatione in symbola conversa.

Sed in novo hoc genere calculi repraesentatoris perpetuo imaginatio mentem vel calamum comitatur, ut quicquid calamus designat, id continue phantasia possit imaginari. Et animus numquam ita symbolis adhaeret, ut rei ipsius considerationem deserere cogatur, ac proinde quaevis lineola est theorema aliquod vel proprietas ipsius figurae vel motus. (Ita tamen ut non sit necesse – nisi velimus – actu ipso calculum imaginatione sequi). Hinc etiam calculo ipso adhibito et constituto sequetur mirifica confirmatio atque confortatio imaginationis, ut possimus in postremum etiam difficillimas res et maxime compositas consequi, idque sine calamo et charta aut figuris, habito semel hoc filo Ariadnaeo mentis in imaginando quod iste calculus nobis ostendet.

Ac jam ostendi in tentamine quodam hujus artis, quomodo sphaerica et plana superficies, itemque circumferentia circuli et linea recta per notas repraesententur exactissime, ita ut eorum quasi imago cum notis menti subjiatur (expressione tamen, non phantasiam tantum, juvante, sed et menti definitiones praebente). Hinc sequitur omnes alias lineas et superficies eodem modo posse exhiberi, quia per has possunt determinari.

Ultima analysis omnium erit in situs seu respectus punctorum. Videndum quomodo cognosci possit punctum quaesitum habere plures valores posibles. Non semper eundem ad ultimam analysin, sed quo licet manendum in mediis, seu in ipsis motuum genesis repraesentationibus; ut in calculis ita et [hic] semper determinanda puncta per locorum intersectiones. Cum magni sit usus calculus Trigonometricus res ita instituenda est ut quaelibet problemata ad ipsum commode referantur, est enim algebraico commodior in praxi.

with all thought being converted into symbols.

But in this new kind of representational calculus, the imagination perpetually accompanies the mind or pen, so that whatever the pen sets down can be imagined by continual mental imagery [*continue phantasia*]. And the mind never so clings to the symbols that it is forced to abandon contemplation of the thing itself, and accordingly every tiny line is some theorem or property of the figure or motion itself, yet so that it is not necessary to actively follow the calculation in the imagination (unless we want to). Hence also, by employing and establishing this calculus, there will follow a wonderful confirmation and revitalization of the imagination. In the end, then, we will be able to follow even the most difficult and complicated things, and that without pen and paper or figures, once we have this Ariadnean thread of the mind in imagining that the calculus will show to us.

And I already showed, in an essay in this art, how a spherical and a planar surface, and likewise the circumference of a circle and a straight line, are represented exactly through marks so that something like an image of them is subjected to the mind (the expression, though, not helping only the imagination [*phantasiam*] but also supplying definitions to the mind). Hence it follows that all other curves and surfaces can be exhibited in the same way, since they can be determined by these.

The final analysis of everything will be into situations, i.e. regarding points. One must see how we can know that a point we seek has multiple possible values. We need not always go to the final analysis, but where possible we should remain in the middle, i.e. in the representations themselves of the generations of motions. As in the calculus, so also here, the points are always to be determined through intersections of loci. Since the trigonometric calculus is of great utility, the thing should be set up such that any problems are conveniently referred to it, for it is more convenient than the algebraic one in practice.